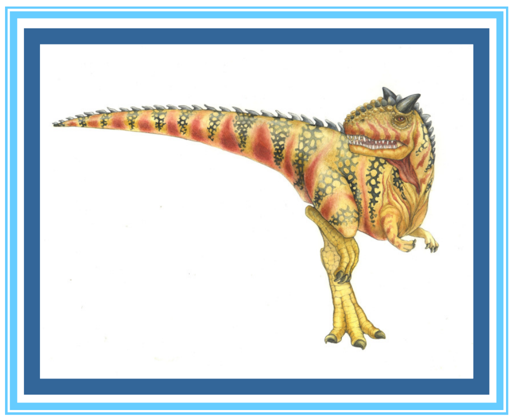
Chapter 19: Network and Distributed Systems



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Chapter 19: Distributed Systems

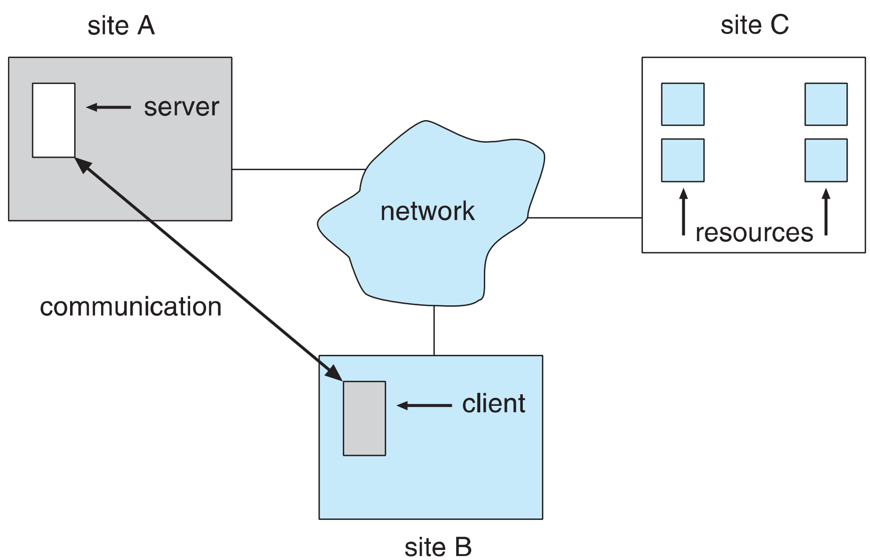
- Advantages of Distributed Systems
- Network Structure
- Communication Structure
- Network and Distributed Operating Systems
- Design Issues of Distributed Systems
- Distributed File Systems

Chapter Objectives

- Explain the advantages of networked and distributed systems
- Provide a high-level overview of the networks that interconnect distributed systems
- Define the roles and types of distributed systems in use today
- Discuss issues concerning the design of distributed file systems

*Overview

- A **distributed system** is a collection of loosely coupled nodes interconnected by a communications network
- Nodes variously called ***processors, computers, machines, hosts***
  - ***Site*** is location of the machine, ***node*** refers to specific system
  - Generally a ***server*** has a resource a ***client*** node at a different site wants to use



*Overview (cont.)

- Nodes may exist in a ***client-server***, ***peer-to-peer***, or ***hybrid*** configuration.
  - In client-server configuration, server has a resource that a client would like to use
  - In peer-to-peer configuration, each node shares equal responsibilities and can act as both clients and servers
- Communication over a network occurs through **message passing**
  - All higher-level functions of a standalone system can be expanded to encompass a distributed system

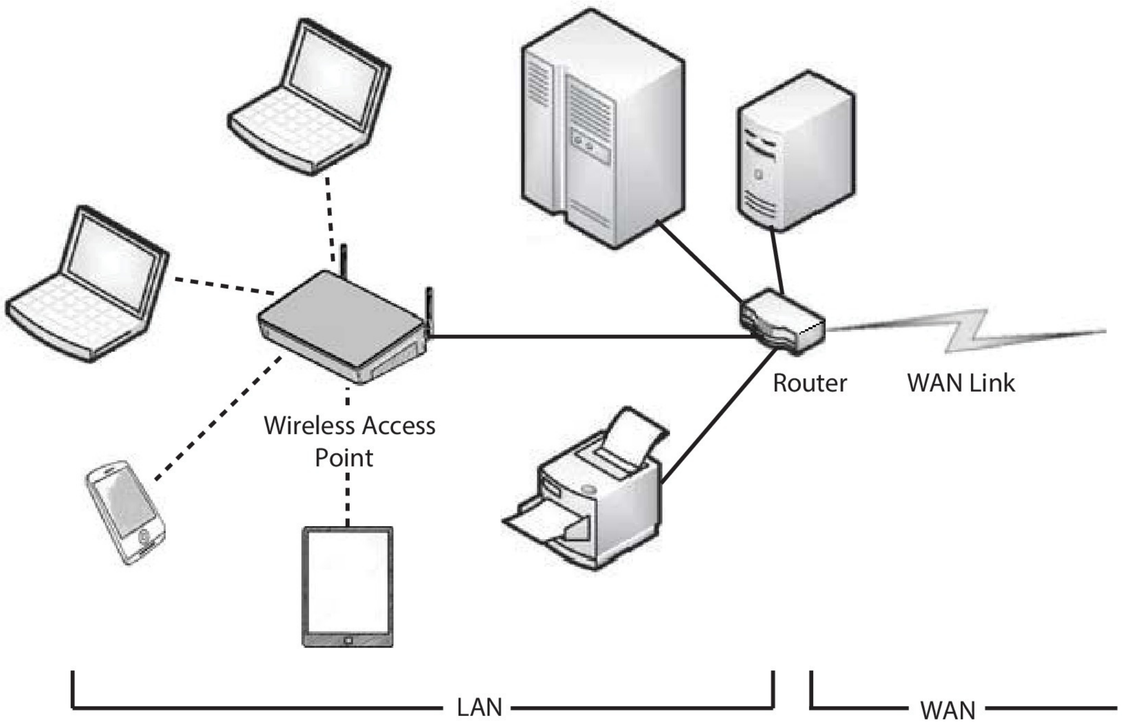
*Reasons for Distributed Systems

- **Resource sharing**
  - Sharing files or printing at remote sites
  - Processing information in a distributed database
  - Using remote specialized hardware devices such as ***graphics processing units*** (GPUs)
- **Computation speedup**
  - Distribute subcomputations among various sites to run concurrently
  - **Load balancing** – moving jobs to more lightly-loaded sites
- **Reliability**
  - Detect and recover from site failure, function transfer, reintegrate failed site

Network Structure

- **Local-Area Network** (**LAN**) – designed to cover small geographical area
  - Consists of multiple computers (workstations, laptops, mobile devices), peripherals (printers, storage arrays), and routers providing access to other networks
  - Ethernet and/or Wireless (**WiFi**) most common way to construct LANs
    - Ethernet defined by standard IEEE 802.3 with speeds typically varying from 10Mbps to over 10Gbps
    - WiFi defined by standard IEEE 802.11 with speeds typically varying from 11Mbps to over 400Mbps.
    - Both standards constantly evolving

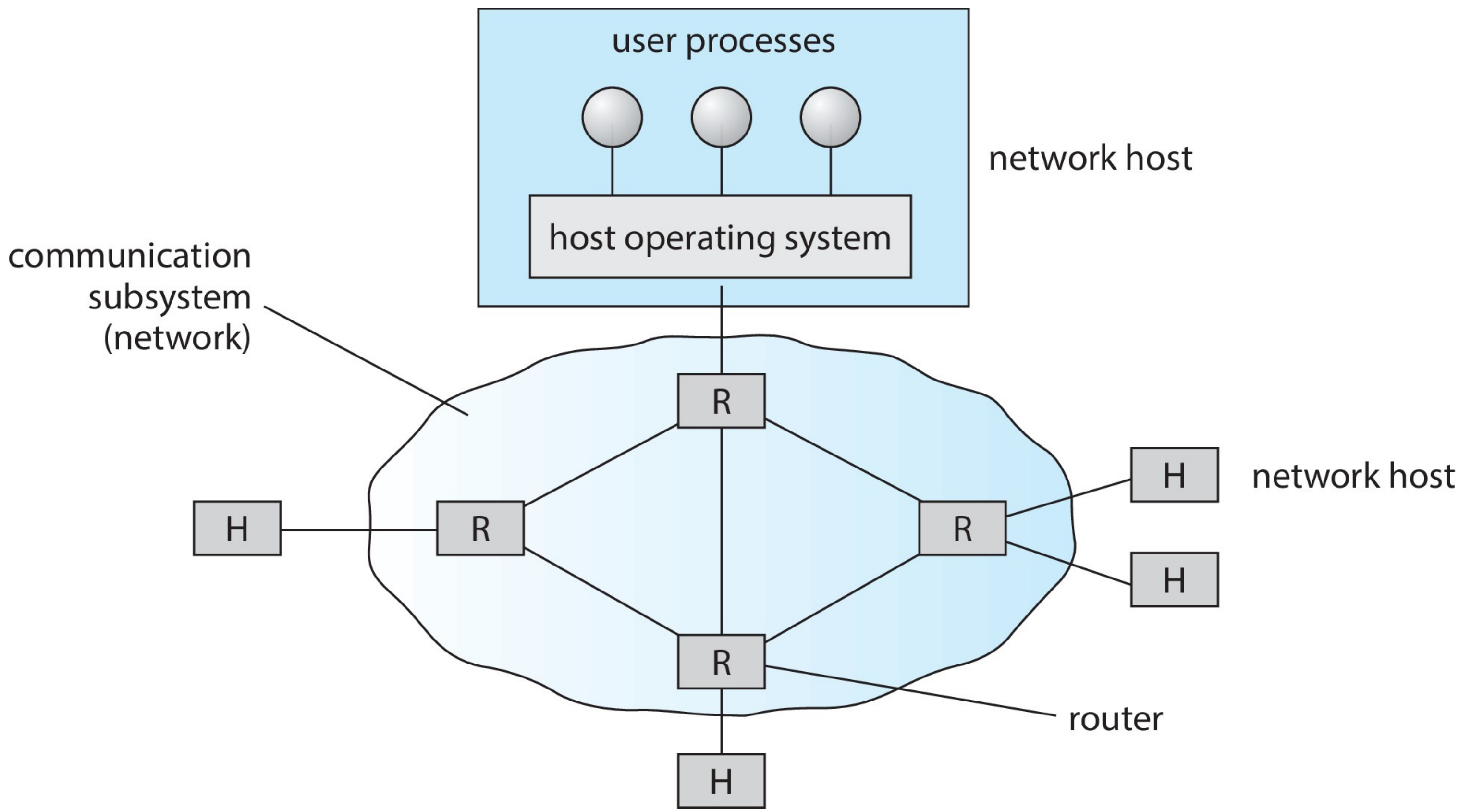
Local-Area Network (LAN)



Network Structure (Cont.)

- **Wide-Area Network** (**WAN**) – links geographically separated sites
  - Point-to-point connections via links
    - Telephone lines, leased (dedicated data) lines, optical cable, microwave links, radio waves, and satellite channels
  - Implemented via **routers** to direct traffic from one network to another
  - Internet (World Wide Web) WAN enables hosts world wide to communicate
  - Speeds vary
    - Many backbone providers have speeds at 40-100Gbps
    - Local **Internet Service Providers** (**ISPs**) may be slower
    - WAN links constantly being upgraded
  - WANs and LANs interconnect, similar to cell phone network:
    - Cell phones use radio waves to cell towers
    - Towers connect to other towers and hubs

Wide-Area Network (WAN)



*Naming and Name Resolution

- Each computer system in the network has a unique name
- Each process in a given system has a unique name (process-id)
- Identify processes on remote systems by

  <**host-name**, **identifier**> pair

- **Domain name system** (**DNS**) – specifies the naming structure of the hosts, as well as name to address **resolution** (Internet)

```
/**
 * Usage: java DNSLookUp <IP name>
 * i.e. java DNSLookUp www.wiley.com
 */
public class DNSLookUp {
   public static void main(String[] args) {
      InetAddress hostAddress;

      try {
         hostAddress = InetAddress.getByName(args[0]);
         System.out.println(hostAddress.getHostAddress());
      }
      catch (UnknownHostException uhe) {
         System.err.println("Unknown host: " + args[0]);
      }
   }
}
```

Figure 19.4 Java program illustrating a DNS lookup.

Communication Protocol

The communication network is partitioned into the following multiple layers:

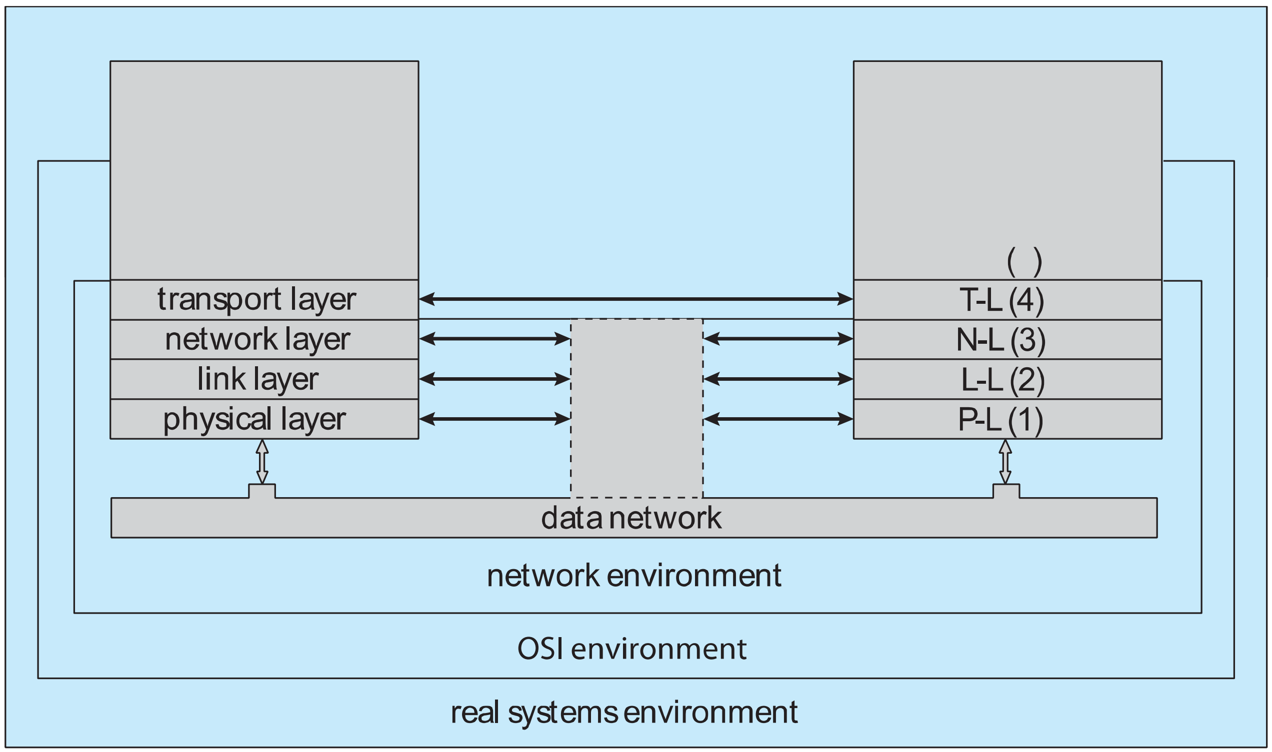
- **Layer 1: Physical layer** – handles the mechanical and electrical details of the physical transmission of a bit stream
- **Layer 2: Data-link layer** – handles the *frames*, or fixed-length parts of packets, including any error detection and recovery that occurred in the physical layer
- **Layer 3: Network layer** – provides connections and routes packets in the communication network, including handling the address of outgoing packets, decoding the address of incoming packets, and maintaining routing information for proper response to changing load levels

Communication Protocol (Cont.)

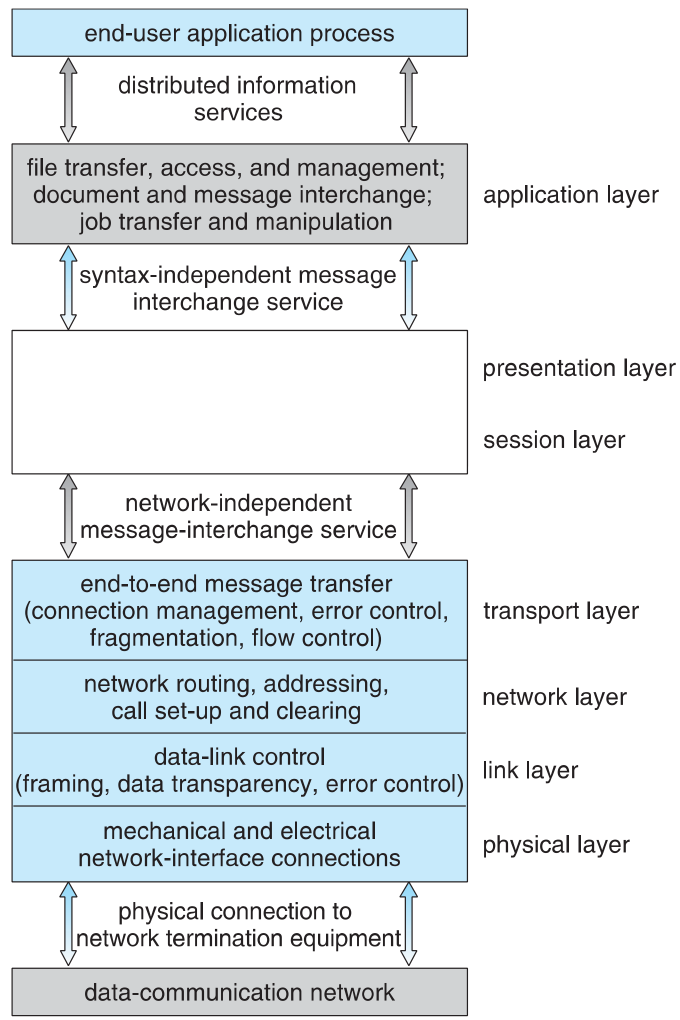
- **Layer 4: Transport layer** – responsible for low-level network access and for message transfer between clients, including partitioning messages into packets, maintaining packet order, controlling flow, and generating physical addresses
- **Layer 5: Session layer** – implements sessions, or process-to-process communications protocols
- **Layer 6: Presentation layer** – resolves the differences in formats among the various sites in the network, including character conversions, and half duplex/full duplex (echoing)
- **Layer 7: Application layer** – interacts directly with the users, deals with file transfer, remote-login protocols and electronic mail, as well as schemas for distributed databases

OSI Network Model

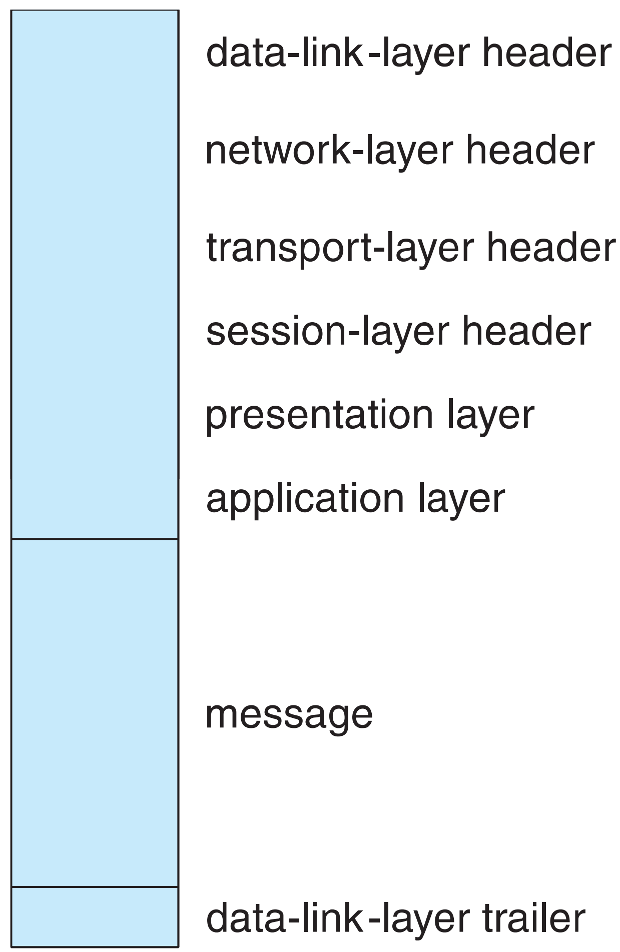
Logical communication between two computers, with the three lowest-level layers implemented in hardware



OSI Protocol Stack



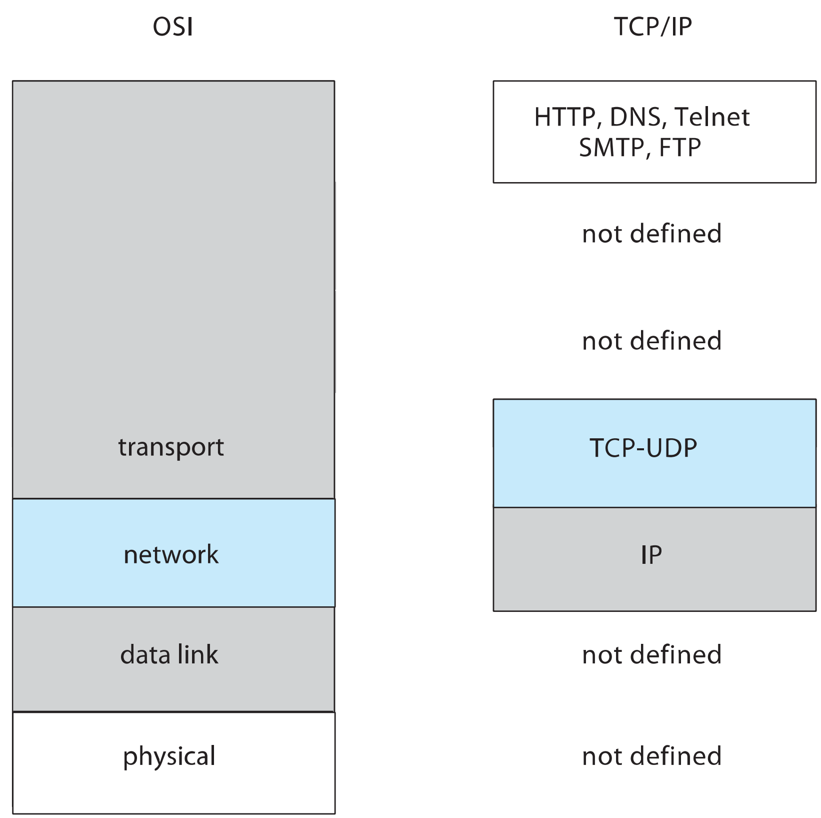
OSI Network Message



The OSI model

- The OSI model formalizes some of the earlier work done in network protocols but was developed in the late 1970s and is currently not in widespread use
- The most widely adopted protocol stack is the TCP/IP model, which has been adopted by virtually all Internet sites
- The TCP/IP protocol stack has fewer layers than the OSI model. Theoretically, because it combines several functions in each layer, it is more difficult to implement but more efficient than OSI networking
- The relationship between the OSI and TCP/IP models is shown in the next slide

The OSI and TCP/IP Protocol Stacks



TCP/IP Example

- Every host has a name and an associated **IP address** (host-id)
  - Hierarchical and segmented
- Sending system checks routing tables and locates a router to send packet
- Router uses segmented network part of host-id to determine where to transfer packet
  - This may repeat among multiple routers
- Destination system receives the packet
  - Packet may be complete message, or it may need to be reassembled into larger message spanning multiple packets

TCP/IP Example (Cont.)

- Within a network, how does a packet move from sender (host or router) to receiver?
  - Every Ethernet/WiFi device has a **medium access control** (MAC) address
  - Two devices on same LAN communicate via MAC address
  - If a system needs to send data to another system, it needs to discover the IP to MAC address mapping
    - Uses **address resolution protocol** (ARP)
  - A broadcast uses a special network address to signal that all hosts should receive and process the packet
    - Not forwarded by routers to different networks

Ethernet Packet

bytes		
7	preamble—start of packet	each byte pattern 10101010
1	start of frame delimiter	pattern 10101011
2 or 6	destination address	Ethernet address or broadcast
2 or 6	source address	Ethernet address
2	length of data section	length in bytes
0–1500	data	message data
0–46	pad (optional)	message must be > 63 bytes long
4	frame checksum	for error detection

Transport Protocols UDP and TCP

- Once a host with a specific IP address receives a packet, it must somehow pass it to the correct waiting process
- Transport protocols TCP and UDP identify receiving and sending processes through the use of a port number
  - Allows host with single IP address to have multiple server/client processes sending/receiving packets
  - ***Well-known*** port numbers are used for many services
    - FTP – port 21
    - ssh – port 22
    - SMTP – port 25
    - HTTP – port 80
- Transport protocol can be simple or can add reliability to network packet stream

User Datagram Protocol

- UDP is ***unreliable*** – bare-bones extension to IP with addition of port number
  - Since there are no guarantees of delivery in the lower network (IP) layer, packets may become lost
  - Packets may also be received out-out-order
- UDP is also ***connectionless*** – no connection setup at the beginning of the transmission to set up state
  - Also no connection tear-down at the end of transmission
- UDP packets are also called **datagrams**

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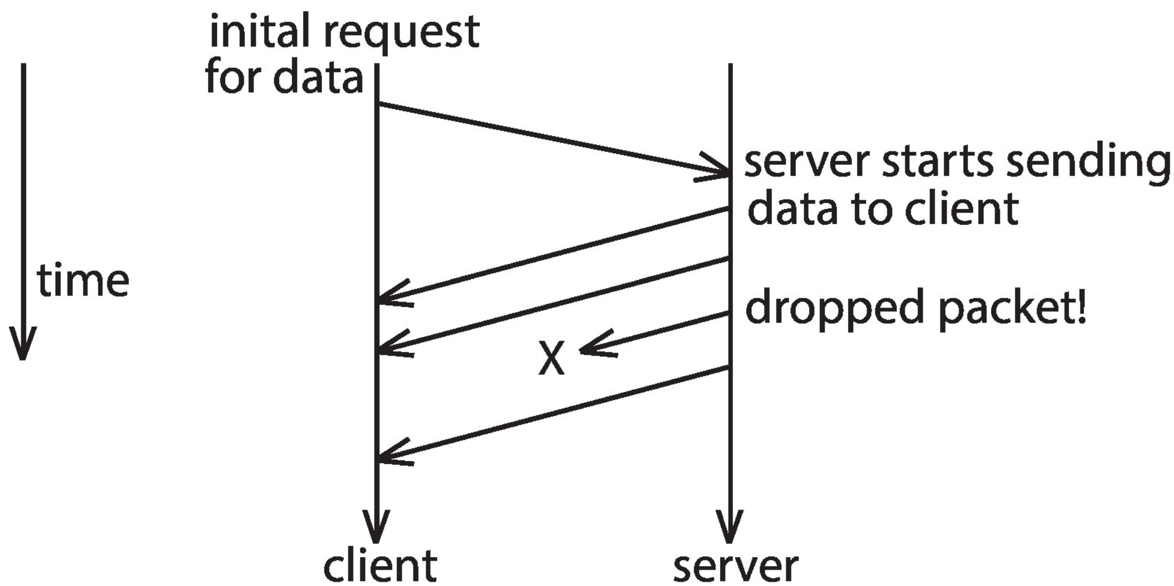
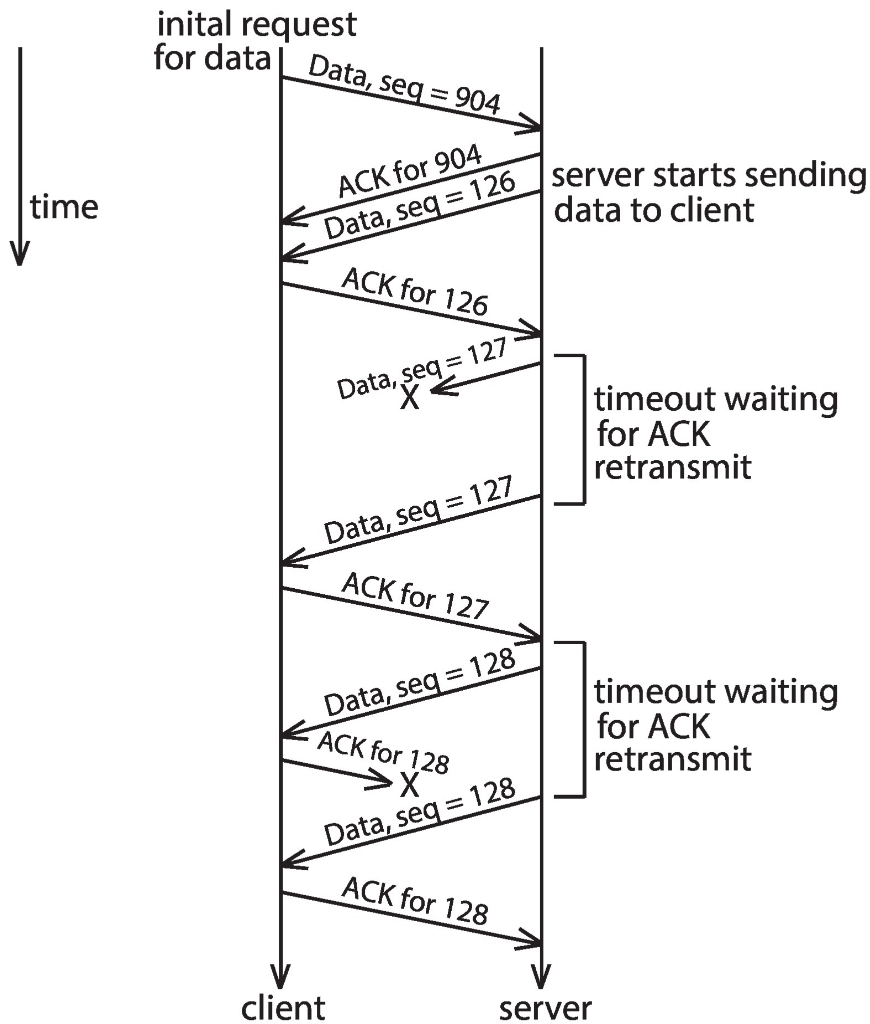
UDP Dropped Packet Example



Transmission Control Protocol

- TCP is both ***reliable*** and ***connection-oriented***
- In addition to port number, TCP provides abstraction to allow in-order, uninterrupted ***byte-stream*** across an unreliable network
  - Whenever host sends packet, the receiver must send an **acknowledgement packet** (ACK). If ACK not received before a timer expires, sender will resend.
  - **Sequence numbers** in TCP header allow receiver to put packets in order and notice missing packets
  - Connections are initiated with series of control packets called a ***three-way handshake***
    - Connections also closed with series of control packets

TCP Data Transfer Scenario



Transmission Control Protocol (Cont.)

- Receiver can send a **cumulative ACK** to acknowledge series of packets
  - Server can also send multiple packets before waiting for ACKs
  - Takes advantage of network throughput
- Flow of packets regulated through **flow control** and **congestion control**
  - ***Flow control*** – prevents sender from overrunning capacity of receiver
  - ***Congestion control*** – approximates congestion of the network to slow down or speed up packet sending rate

*Network-oriented Operating Systems

- Two main types
- **Network Operating Systems**
  - Users are aware of multiplicity of machines
- **Distributed Operating Systems**
  - Users not aware of multiplicity of machines

*Network Operating Systems

- Users are aware of multiplicity of machines
- Access to resources of various machines is done explicitly by:
  - Remote logging into the appropriate remote machine (ssh)
    - `ssh kristen.cs.yale.edu`
  - Transferring data from remote machines to local machines, via the File Transfer Protocol (FTP) mechanism
  - Upload, download, access, or share files through cloud storage
- Users must change paradigms – establish a **session**, give network-based commands, use a web browser
  - More difficult for users

*Distributed Operating Systems

- Users not aware of multiplicity of machines
  - Access to remote resources similar to access to local resources
- **Data Migration** – transfer data by transferring entire file, or transferring only those portions of the file necessary for the immediate task
- **Computation Migration** – transfer the computation, rather than the data, across the system
  - Via remote procedure calls (RPCs)
  - Via messaging system

*Distributed-Operating Systems

- **Process Migration** – execute an entire process, or parts of it, at different sites
  - ***Load balancing*** – distribute processes across network to even the workload
  - ***Computation speedup*** – subprocesses can run concurrently on different sites
  - ***Hardware preference*** – process execution may require specialized processor
  - ***Software preference*** – required software may be available at only a particular site
  - ***Data access*** – run process remotely, rather than transfer all data locally
- Consider the World Wide Web

*Design Issues of Distributed Systems

- We investigate three design questions:
  - ***Robustness*** – Can the distributed system withstand failures?
  - ***Transparency*** – Can the distributed system be transparent to the user both in terms of where files are stored and user mobility?
  - ***Scalability*** – Can the distributed system be scalable to allow addition of more computation power, storage, or users?

*Distributed File System

- **Distributed file system** (**DFS**) – a file system whose clients, servers, and storage devices are dispersed among the machines of a distributed system
  - Should appear to its clients as a conventional, centralized file system
- Key distinguishing feature is management of dispersed storage devices

*Distributed File System (Cont.)

- **Service** – software entity running on one or more machines and providing a particular type of function to a priori unknown clients
- **Server** – service software running on a single machine
- **Client** – process that can invoke a service using a set of operations that forms its client interface
- A client interface for a file service is formed by a set of primitive file operations (create, delete, read, write)
- Client interface of a DFS should be transparent; i.e., not distinguish between local and remote files
- Sometimes lower level **inter-machine** interface need for cross-machine interaction

*Distributed File System (Cont.)

- Two widely-used architectural models include **client-server model** and **cluster-based model**
- Challenges include:
  - Naming and transparency
  - Remote file access
  - Caching and cache consistency

*Naming and Transparency

- **Naming** – mapping between logical and physical objects
- **Multilevel mapping** – abstraction of a file that hides the details of how and where on the disk the file is actually stored
- A **transparent** DFS hides the location where in the network the file is stored
- For a file being **replicated** in several sites, the mapping returns a set of the locations of this file's replicas; both the existence of multiple copies and their location are hidden

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*Remote File Access

- Consider a user who requests access to a remote file. The server storing the file has been located by the naming scheme, and now the actual data transfer must take place.
- **Remote-service mechanism** is one transfer approach.
  - A requests for accesses are delivered to the server, the server machine performs the accesses, and their results are forwarded back to the user.
  - One of the most common ways of implementing remote service is the RPC paradigm

*Remote File Access (Cont.)

- Reduce network traffic by retaining recently accessed disk blocks in a cache, so that repeated accesses to the same information can be handled locally
  - If needed data not already cached, a copy of data is brought from the server to the user
  - Accesses are performed on the cached copy
  - Files identified with one master copy residing at the server machine, but copies of (parts of) the file are scattered in different caches
- **Cache-consistency problem** – keeping the cached copies consistent with the master file
  - Could be called **network virtual memory**

End of Chapter 19